

R Series: Caret Package

In this 25th article in the 'R, Statistics and Machine Learning' series, we explore the caret package, which is a framework for building machine learning models in R.



The caret package provides miscellaneous functions for training classification and regression models. It allows you to tune parameters, use resampling techniques and make predictions from the models.

We will use R version 4.2.2 installed on Parabola GNU/Linux-libre (x86-64) for the R code snippets.

```
$ R --version
```

```
R version 4.2.2 (2022-10-31) -- "Innocent and Trusting"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

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Package installation

You can install and load the caret package using the following commands:

```
> install.packages("caret")
```

```
Installing package into '/home/shakthi/R/x86_64-pc-linux-gnu-
library/4.1'
```

```
...
```

```
* building package indices
* installing vignettes
* testing if installed package can be loaded from temporary
location
* checking absolute paths in shared objects and dynamic
libraries
* testing if installed package can be loaded from final
location
* testing if installed package keeps a record of temporary
installation path
DONE (caret)
```

```
> library(caret)
Loading required package: ggplot2
Loading required package: lattice
```

We will install and use a few additional packages to demonstrate the use of the *caret* package. The *earth* package implements multivariate adaptive regression splines (MARS). You can install and load this library using the following commands: